

Solution Requirements

Date	01 july 2026
Team ID	
Project Name	Smart Lender – AI Powered Loan Eligibility Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system shall allow users to securely register, log in, and log out using their email and password. User sessions shall be maintained securely.	High
2	Authorization Levels	The system shall allow authenticated users to access loan prediction, dashboard, prediction history, and PDF reports. Unauthorized users cannot access protected pages.	High
3	External Interfaces	The application shall integrate with the trained Random Forest Machine Learning model, SQLite database, and ReportLab library for PDF generation.	High
4	Transaction Processing	The system shall receive applicant details, process them using the Random Forest model, generate predictions, and store results in the database.	High
5	Reporting	The system shall generate prediction reports, display dashboard analytics, maintain prediction history, and allow users to download PDF reports.	High
6	Business Rules	Loan eligibility shall be predicted based on applicant details such as income, education, employment status, credit history, loan amount, and property area using the trained Random Forest model.	High

7	Compliance to Laws or Regulations	The application shall protect user information, securely store credentials, and follow standard data privacy and security practices.	Medium
8	Other	The system shall provide a responsive user interface, prediction history management, dashboard analytics, and printable reports.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate loan predictions and load dashboard pages quickly.	Average response time less than 2 seconds.
2	Scalability	The system should support future migration to cloud databases and handle increasing numbers of users and prediction records.	Supports future deployment for 5000+ users.
3	Security & Data Privacy	User passwords shall be securely stored, sessions managed safely, and unauthorized access prevented.	Password hashing, secure sessions, protected routes.
4	Reliability & Availability	The application should remain available during normal operation with minimal downtime.	99% uptime during deployment.

5	Usability & Accessibility	The user interface shall be simple, responsive, and easy to use on desktops, tablets, and mobile devices.	Responsive design with user-friendly navigation
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